

Lab 4: DECISION TREE & RANDOM FOREST

DUE: 13:00 15/05/2026

Submission: mlearning

- **Lab#_YOURNAME.ipynb**
- Must include all data file (**.csv**,...) and **.ipynb** in your submission
- **.csv** and **.ipynb** must be in the **same** folder (`pd.read_csv('data.csv')`)
- No use of LOCAL libraries/folders
- NO LATE submission accepted

Input: **dataset.csv**

Use the **same** testing set for the following questions:

1. Use Decision Tree (DT) (gini index) to detect faults in a power system. Compute the accuracy, precision, recall, F1
2. Now, use Random Forest (RF) to detect faults in a power system. Compare the performances of RF from using different trees: 10 trees, 100 trees and 500 trees. Does the accuracy improve when more DT are used? Compute the RF accuracy, precision, recall, F1.
3. Compare DT and RF. Does RF outperform DT?
4. Now, use 100 trees, but **max_depth**, the maximum depth of the tree, is restricted to **3**. Compute the RF accuracy.
5. Feature importance: Use `RandomForestClassifier.feature_importances_` to rank features of the dataset (For more information: https://scikit-learn.org/stable/auto_examples/ensemble/plot_forest_importances.html)
6. Now drop the **least** important feature. Train RF model and compute the accuracy. Is there any improvement?